Real Analysis: Answer to Exercise 2

Due date:

Question 1. **Solution**: Let $f, g : \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = x^2, \qquad g(x) = \sin x \quad \forall x \in \mathbb{R}.$$

Then

$$f \circ g(x) = \sin^2(x) \neq \sin(x^2) = g \circ f(x).$$

The functions f and g therefore meet the requirements of the question.

Question 2. **Solution**: Recall that $f : X \to Y$ is injective if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for all $x_1, x_2 \in X$. Suppose $f(x_1) = f(x_2)$ for some $x_1, x_2 \in X$. Then

$$x_1 = g(f(x_1)) = g(f(x_2)) = x_2,$$

so f is injective.

For surjectivity, recall that $g : Y \to X$ is surjective if for every $x \in X$ there exists $y \in Y$ with $g(y) = x$. Let $x \in X$. Then $g(f(x)) = x$; setting $y = f(x)$ gives $g(y) = x$. Hence g is surjective.

Question 3. **Solution**: It is possible. For example, let $f : \{1, 2\} \to \{1, 2, 3\}$ and $g : \{1, 2, 3\} \to \{1, 2\}$ be given by

$$f(x) = \begin{cases} 1, & x = 1, \\ 2, & x = 2, \end{cases} \qquad g(x) = \begin{cases} 1, & x = 1, \\ 2, & x = 2, \\ 2, & x = 3. \end{cases}$$

Then f is not surjective, since $f(\{1, 2\}) = \{1, 2\} \neq \{1, 2, 3\}$, and g is not injective, since $g(2) = g(3) = 2$; so neither f nor g is a bijection. However,

$$g \circ f(x) = \begin{cases} 1, & x = 1, \\ 2, & x = 2, \end{cases}$$

so $g \circ f : \{1, 2\} \to \{1, 2\}$ is a bijection, as required.

Question 4. **Solution**: For (1), if $y \in f(C \cup D)$, then there is $x \in C \cup D$ such that $f(x) = y$. If $x \in C$, then $y = f(x) \in f(C)$; if $x \in D$, then $y = f(x) \in f(D)$. Hence $f(C \cup D) \subset f(C) \cup f(D)$. Conversely, $f(C) \subset f(C \cup D)$ and $f(D) \subset f(C \cup D)$, so $f(C) \cup f(D) \subset f(C \cup D)$. Therefore $f(C \cup D) = f(C) \cup f(D)$.

For (2), if $y \in f(C \cap D)$, there is $x \in C \cap D$ with $y = f(x)$; since $x \in C$ and $x \in D$, we have $y \in f(C)$ and $y \in f(D)$, so $y \in f(C) \cap f(D)$. Hence $f(C \cap D) \subset f(C) \cap f(D)$.

For the example, let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, with $C = (0, \infty)$ and $D = (-\infty, 0)$. Then $f(C \cap D) = f(\varnothing) = \varnothing$, while $f(C) \cap f(D) = (0, \infty) \cap (0, \infty) = (0, \infty)$, so $f(C \cap D) \neq f(C) \cap f(D)$.

Question 5. **Solution**: For (a), $x \in f^{-1}(C \cup D) \iff f(x) \in C \cup D \iff f(x) \in C$ or $f(x) \in D \iff x \in f^{-1}(C)$ or $x \in f^{-1}(D) \iff x \in f^{-1}(C) \cup f^{-1}(D)$.

For (b), $x \in f^{-1}(C \cap D) \iff f(x) \in C \cap D \iff f(x) \in C$ and $f(x) \in D \iff x \in f^{-1}(C) \cap f^{-1}(D)$.

For (c), $x \in f^{-1}(C^c) \iff f(x) \in C^c \iff f(x) \notin C \iff x \notin f^{-1}(C) \iff x \in \big(f^{-1}(C)\big)^c$.

Question 6. **Solution**: (a) Consider the open intervals

$$A_n = \left(-1 - \frac{1}{n}, 1 + \frac{1}{n}\right), \qquad \text{so that} \qquad \bigcap_{n=1}^{\infty} A_n = [-1, 1],$$

which is closed, and hence not open.

(b) Consider the closed intervals

$$B_n = \left[-1 + \frac{1}{n}, 1 - \frac{1}{n}\right], \qquad \text{so that} \qquad \bigcup_{n=1}^{\infty} B_n = (-1, 1),$$

which is open, and hence not closed.